

THE GLOBAL SOUTH SAS

GSDC Platform

CTO Architecture & Operations Guide

v4.0 — Post-Session Master Reference

May 2026 · Confidential · For CTO, Dev Lead, Engineering Team

Today's Session — What We Built

In a single extended CTO architecture session on 5 May 2026, we took the GSDC settlement system from a document with four unresolved open questions — chain architecture undecided, procurement pathway unclear, margin rates unconfirmed, Partner Platform PRD not yet written — to a fully specified, signed-off engineering foundation ready to build. Starting from the v2 Settlement System memo, we resolved the tech stack (EVM / Solidity / Ethereum / Sepolia testnet), selected Polygon CDK with Gateway.fm as the production chain strategy, evaluated and rejected Base on sovereignty grounds, designed the x402 modular integration architecture, mapped the MCP AI-native layer, specified and gap-analysed the PartnerAdapter Interface through two iterations to a final v1.2 sign-off, built a jurisdiction-agnostic multi-tenant Partner Platform PRD informed by live research into BCB Resolution 521 and HKMA MSO requirements, and produced eight production-quality documents. The platform is now fully specified at the architecture and product layer. The engineering team can build without ambiguity.

Documents produced: 8 — CTO Architecture Guide v4 · Roadmap v4 · Settlement CTO Response · PartnerAdapter v1.0/1.1/1.2 · Partner Platform PRD v1.0/1.1 · Sign-Off Record

Decisions closed: 12 major architecture decisions resolved from open to locked

Status: PartnerAdapter Interface Spec v1.2: SIGNED OFF · Partner Platform PRD v1.1: SIGNED OFF · Build: READY TO START

1. Session Brief — 5 May 2026

What we set out to do

Review the v2 Settlement System memo, close all open architectural questions, and produce a complete, buildable specification for the GSDC settlement platform.

1.1 Twelve decisions made today

#	Decision	Resolution
D 1	Tech stack	EVM · Solidity · Ethereum. Sepolia testnet for all Phase 1. Mainnet = Ethereum (GSDC already deployed). Chain selection (Besu vs Polygon CDK) deferred to pre-mainnet milestone.
D 2	x402 integration approach	Modular. EIP-3009 in Phase 1. Full x402 facilitator built in Phase 4 after pilot stable. Agents-as-a-Service dropped.
D 3	GSDC x402 asset type	Native — GSDC implements EIP-3009 directly. Same standing as USDC in the x402 ecosystem.
D 4	Chain sovereignty rule	Base ruled out — US-controlled, Coinbase-operated, USDC-privileging. Incompatible with GSDC self-reliance principle. Polygon CDK (Singapore HQ) confirmed as production chain path.
D 5	Production chain strategy	GSDC Settlement Chain on Polygon CDK via AggLayer. Gateway.fm (EU) as RaaS partner for pilot proving. In-house sequencer and prover migration at scale.
D 6	ZK prover approach	Managed (Gateway.fm) for pilot at ~\$5–8/month at pilot volume. Self-hosted cloud (Hetzner EU) at scale. On-prem at full corridors. Prover must be brought in-house on same timeline as sequencer.
D 7	GSDC procurement (pilot)	Pathway D — domestic OTC via Brazilian-licensed VASP. Lowest IOF risk, entire transaction within Brazil. Pathway C (private pool) built in Phase 2 in parallel.
D 8	MCP integration	GSDC MCP Server as Phase 4 component. Exposes 4 settlement tools to any MCP-compatible AI agent. Built on top of x402 facilitator. <code>get_quote</code> · <code>confirm_settlement</code> · <code>get_float_balance</code> · <code>list_settlements</code> .
D 9	Safe Harbour VM	Not a replacement for Polygon CDK. Relevant as execution environment for off-chain components (Settlement Orchestrator, Fiat Gateway). Deferred evaluation — does not affect on-chain architecture.
D 10	ComplianceCheck third state	passed: boolean "review". Review triggers suspicious transaction queue without blocking settlement. SANCTIONS_LIST cannot return review — hard block only.
D 11	complianceFields on settlement types	Record<string, string> added to QuoteRequest and SettlementRecord. Keys driven by Corridor Profile. Jurisdiction-agnostic adapter, jurisdiction-specific data.
D 12	Partner Platform architecture	Multi-tenant platform. TGS = operator. Each licensed institution = independent tenant. Corridor Profiles drive all

		jurisdiction-specific behaviour. Adding new jurisdiction = new Corridor Profile only.
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2. System Architecture Overview

2.1 The two-layer system

The GSDC platform has two distinct layers. The Partner Platform is the web-facing interface through which licensed local partners interact with the settlement infrastructure. The Settlement Layer is the on-chain infrastructure that executes the actual atomic transfer. Both are operated by TGS from Uruguay.

2.2 Three-actor model

Actor	Institution	Role
LP-BR	Licensed Brazilian institution	Collects BRL from importer. Performs KYC/AML. Triggers settlement via Partner Platform. Holds GSDC float wallet.
TGS Uruguay	The Global South SAS	Issues GSDC. Operates Partner Platform and Settlement Layer. Runs DON+DAO oracle. Acts as router — not custodian, not FX dealer.
LP-HK	Licensed HK institution	Receives GSDC on-chain. Delivers CNH to Chinese end-user from pre-funded working capital. Holds GSDC float wallet.

2.3 Six-layer architecture

Layer	Component	Description
L1	Oracle network	DON+DAO — Decentralised Oracle Network with DAO governance. EIP-712 signed reports. Median aggregation. 5-min TTL. Beta complete, running on interim VMs.
L2	Smart contracts	Solidity / EVM. GSDC ERC-20 + EIP-3009. Settlement Diamond (9 facets). Margin Wallet Contracts. Compliance Gate. Deployed to Sepolia testnet.
L3	Settlement Orchestrator	Quote Engine · Settlement State Machine · idempotency · retry logic · failure recovery. Node.js/TypeScript. PartnerAdapter interface v1.2.
L4	Partner Platform	Track A: Hosted Dashboard (pilot default). Track B: Full API + webhooks (post-pilot). Multi-tenant. Corridor Profiles. Jurisdiction compliance modules.
L5	Corridor adapters	BR Adapter (PIX · BCB 519/520/521 · IOF). HK Adapter (CIPS/SWIFT · HKMA · MSO). x402 Facilitator (Phase 4). MCP Server (Phase 4).
L6	Partner integration	Configuration only. Requires signed partner. LP-BR config (PIX key · BCB participant code · Receita Federal ID). LP-HK config (CIPS code · HKMA entity ID · MSO licence).

2.4 Technology stack — all layers

Component	Technology	Status
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GSDC ERC-20 token	Solidity · Ethereum mainnet · CertiK audited Dec 2025	LIVE on Ethereum mainnet
EIP-3009 extension	Solidity · same contract · same audit cycle	Phase 1 build — Week 1
Settlement Contract (Diamond)	Solidity · EIP-2535 · 9 facets · Sepolia	Phase 1 build — Week 1-6
DON+DAO Oracle	EIP-712 signed · median aggregation · interim VMs	Beta complete · production migration in progress
Settlement Orchestrator	Node.js · TypeScript · PartnerAdapter v1.2	Phase 1 build
Partner Platform (Track A)	Multi-tenant web app · Corridor Profiles · i18n	Phase 1 build — from PRD freeze
Testnet / RPC	Ethereum Sepolia · Chainstack EU (non-US)	Ready to set up — Week 1
Production chain	Polygon CDK appchain via AggLayer · Gateway.fm RaaS	Phase 2-3 · deferred from Phase 1
x402 Facilitator	@gsdc/x402-facilitator · EIP-3009 · CAIP-2 · 3 HTTP endpoints	Phase 4 — after pilot stable
MCP Server	@gsdc/mcp-server · 4 tools · wraps Settlement Orchestrator	Phase 4 — after x402 facilitator
Cloud infrastructure	Non-US only: Hetzner EU / OVH / Contabo	Setup Week 1

3. Complete Component Inventory

Every module in the system classified by ring, layer, phase, and current build status.

Module	Class	Layer	Phase	Status	Notes
GSDC ERC-20 Token	CORE	L2	LIVE	done	CertiK audited Dec 2025. Deployed Ethereum mainnet.
EIP-3009 (Transfer With Authorization)	CORE	L2	P1	ready	Add to existing token contract. Same audit cycle as Settlement Contract. Prerequisite for x402.
DON+DAO Oracle Contract	CORE	L1	P1	inprog	Beta complete. Running on interim VMs. Oracle contract to deploy to Sepolia. IT infra decision made — VMs confirmed.
Settlement Contract — 9-facet Diamond	CORE	L2	P1	ready	Quote Verifier · Float Manager · Settlement Executor · Margin Splitter · Compliance Gate · Time-Lock · Dispute Resolver · Event Emitter · Mint/Burn Authority.
Margin Wallet Contracts	CORE	L2	P1	ready	Per-partner accumulation and withdrawal. Same build cycle as Settlement Contract.
PartnerAdapter Interface Spec v1.2	CORE	L3	P1	done	SIGNED OFF. complianceFields added. Three-state ComplianceCheck. All 18 gaps resolved.
MockAdapter	CORE	L3	P1	ready	Implements full PartnerAdapter interface v1.2 with fake fiat rails. Unblocked by sign-off.
Quote Engine	CORE	L3	P1	ready	Corridor + amount → oracle rate → fee stack → signed quote + TTL. Needs PartnerAdapter freeze (done) + Oracle live.
Settlement State Machine	CORE	L3	P1	ready	6 states + idempotency + retries + failure recovery. Needs PartnerAdapter freeze (done).
Audit & Reconciliation Layer	CORE	L4	P1	ready	On-chain event indexer · ledger sync · mismatch alert · corridor freeze.

Partner Platform — Track A (Hosted Dashboard)	CORE	L4	P1	ready	PRD v1.1 SIGNED OFF. Multi-tenant · Corridor Profiles · Jurisdiction compliance modules. Build starts from PRD freeze.
Partner Platform — Track B (Full API)	CORE	L4	P3	deferred	Post-pilot. Separate PRD to be written after pilot stable.
Corridor Profile Config API	CORE	L4	P1	ready	Serves Corridor Profile data to dashboard at session init. Supports Jurisdiction-agnostic architecture.
BR Adapter (generic → configured)	ADAPTER	L5	P2	ready	PIX · BCB 519/520/521 · IOF · eFX fields. Draft from Week 8.
HK Adapter (generic → configured)	ADAPTER	L5	P2	ready	CIPS/SWIFT · HKMA MSO · suspicious tx queue. Draft from Week 8.
x402 Facilitator (@gsdc/x402-facilitator)	ADAPTER	L5	P4	deferred	POST /verify · POST /settle · GET /supported. Reuses @x402/evm types. Requires EIP-3009 live.
x402 Corridor Adapter module	ADAPTER	L5	P4	deferred	Routes x402 micropayments to Settlement Engine.
x402 Session Manager	ADAPTER	L5	P4	deferred	V2 wallet-based identity. Reduces handshake latency for repeat callers.
MCP Server (@gsdc/mcp-server)	ADAPTER	L5	P4	deferred	4 tools: get_quote · confirm_settlement · get_float_balance · list_settlements.
x402 Audit Feed extension	ADAPTER	L4	P4	deferred	Micropayment events indexed alongside settlement events.
LP-BR Integration config	PARTNER	L6	P2	blocked	Config only. Requires signed LP-BR partner.
LP-HK Integration config	PARTNER	L6	P2	blocked	Config only. Requires signed LP-HK partner.

Status key	Meaning
done	Complete and signed off. No further changes without CTO approval.
inprog	Build in progress. Oracle beta running on interim VMs.
ready	Ready to build. All prerequisites met or will be met this week.
deferred	Scheduled for a future phase. Does not block current work.
blocked	Blocked on external dependency (partner contract, legal opinion).

4. Master Todo List — All Phases

This Week — Starts Immediately

#	Task	Owner	Why unblocked
1	Set up Sepolia RPC via Chainstack EU	Sandip	Tech stack confirmed. Non-US provider. Start today.
2	Deploy Diamond scaffold (EIP-2535) to Sepolia	Sandip	Reference implementation exists. Configuration only.
3	Commit PartnerAdapter TypeScript interface file to repo	Dev Lead	Spec v1.2 signed off. This is the build artefact from the spec.
4	Begin MockAdapter implementation against v1.2 interface	Dev Lead	Unblocked by sign-off. 1 week.
5	Scope and start EIP-3009 on GSDC token contract	Smart contract dev	One well-defined interface addition. Same audit cycle.
6	Begin Partner Platform project scaffold + auth + i18n	Frontend + backend	PRD v1.1 signed off. Scaffold, 2FA, i18n framework start now.
7	Identify Pathway D OTC counterparty in Brazil	CEO / COO	Required before pilot go-live. No engineering dependency.
8	Chase Brazilian counsel IOF opinion	Legal / COO	Gates Pathway C decision and off-ramp cost model.
9	Board confirmation of 0.5/0.5/0.5 margin rates	COO / CEO	Required before any partner term sheet is signed.

Phase 1 — Core Build (Weeks 1–14)

ID	Task	Owner	Effort	Dependency
1.1	Sepolia RPC + dev environment (Chainstack EU)	Sandip	1 day	Nothing
1.2	Deploy Diamond scaffold to Sepolia	Sandip	1 day	1.1
1.3	EIP-3009 on GSDC token contract	Smart contract dev	1 wk	1.1
1.4	Settlement Contract — 9-facet Diamond build and deploy to Sepolia	Sandip	4–6 wk	1.1 · v1.2 spec
1.5	Margin Wallet Contracts	Smart contract dev	incl. 1.4	1.1
1.6	CertiKit audit submission (Settlement Contract + EIP-3009)	CTO	—	1.4 1.5 code freeze
1.7	Oracle aggregator contract deploy to Sepolia (DON+DAO)	Oracle team	2–3 wk	1.1 · interim VMs (done)

1.8	Quote Engine (corridor + amount → oracle → fee → quote + TTL)	Dev Lead / Sandip	3–4 wk	v1.2 spec · 1.7
1.9	Settlement State Machine (6-state · idempotency · retries)	Dev Lead / Sandip	5–7 wk	v1.2 spec · 1.8
1.1 0	Commit PartnerAdapter v1.2 TypeScript interface to repo	Dev Lead	Day 1	Sign-off (done)
1.1 1	MockAdapter full implementation against v1.2	Dev Lead	1 wk	1.10
1.1 2	Corridor Profile Config API	Backend dev	2 wk	v1.2 spec
1.1 3	Partner Platform — project scaffold + auth + 2FA + i18n	Frontend + backend	2 wk	PRD v1.1 (done)
1.1 4	Partner Platform — multi-tenancy data model + RBAC	Backend dev	2 wk	PRD v1.1 (done)
1.1 5	Partner Platform — home screen + settlement feed	Frontend	3 wk	1.13 1.14
1.1 6	Partner Platform — quote request + dual-approval flow + compliance fields	Frontend	4 wk	1.13 1.14 1.12
1.1 7	Partner Platform — float management + margin wallet	Frontend	2 wk	1.13 1.14
1.1 8	Partner Platform — reconciliation + BCB daily report generator	Frontend + backend	3 wk	1.14
1.1 9	Partner Platform — HK MSO compliance module	Frontend + backend	2 wk	1.14
1.2 0	Partner Platform — notification system	Backend	2 wk	1.14
1.2 1	Audit and Reconciliation Layer (event indexer · ledger sync · mismatch alert)	Backend dev	3–4 wk	v1.2 spec
1.2 2	End-to-end testnet: all contracts + orchestrator + MockAdapter both sides	QA + Dev Lead	1 wk	1.4 1.8 1.9 1.21
1.2 3	EIP-3009 security review (CertiK)	Smart contract dev + CertiK	incl. 1.6	1.3

Phase 2 — Adapters (Weeks 8–18)

ID	Task	Owner	Effort	Dependency
2.1	Draft generic BR Adapter spec (PIX · BCB · IOF · eFX · complianceFields)	Dev Lead + legal	1 wk	v1.2 spec

2.2	Draft generic HK Adapter spec (CIPS · HKMA · MSO · suspicious tx)	Dev Lead + legal	1 wk	v1.2 spec
2.3	Build BR Adapter generic implementation	Backend dev	3 wk	2.1
2.4	Build HK Adapter generic implementation	Backend dev	3 wk	2.2
2.5	Testnet: BR Adapter (real) + HK MockAdapter	QA + Dev Lead	1 wk	2.3 1.22
2.6	Testnet: BR MockAdapter + HK Adapter (real)	QA + Dev Lead	1 wk	2.4 1.22
2.7	Lock LP-BR partner-specific config on signing	Dev Lead + LP-BR	—	2.3 · partner signed
2.8	Lock LP-HK partner-specific config on signing	Dev Lead + LP-HK	—	2.4 · partner signed
2.9	Onboard LP-BR ops team to Track A staging	Product + LP-BR	1 wk	1.22 2.7
2.10	Onboard LP-HK ops team to Track A staging	Product + LP-HK	1 wk	1.22 2.8
2.11	Pathway C private pool engineering begins (parallel)	Smart contract dev	4 mo	IOF counsel · 2.1

Phase 3 — Pilot Live (Weeks 18–26)

ID	Task	Owner	Effort	Dependency
3.1	Go-live: first BR→CN settlement (constrained volume, manual approval each)	CTO + ops	—	2.7 2.8 1.6 audit
3.2	Monitor: audit layer watching every settlement for discrepancy	Ops + Dev Lead	ongoing	3.1
3.3	Confirm corridor economics over 4–8 weeks live volume	COO + CTO	—	3.1
3.4	Partner technical discovery calls (HashKey / FOMO Pay / StraitsX)	CMO / CEO	1–2 mo	—
3.5	Polygon CDK engagement — confirm Gateway.fm contract and chain setup timeline	CTO	—	3.3
3.6	Build Partner API + webhooks (Track B) — begins after pilot stable	Backend dev	6–8 wk	3.3
3.7	Draft +N corridor adapter specs (IN, ZA, TH, ID) — generic profiles	Dev Lead	ongoing	3.3 v1.2

3.8	Phase 4 prerequisites checklist: EIP-3009 live · pilot volume baseline · no open contract issues	CTO	—	3.3 1.23
3.9	Phase 4 go/no-go decision	CTO + CEO	—	3.8

Phase 4 — x402 + MCP Modular Layer (After pilot stable)

ID	Task	Owner	Effort	Dependency
4.1	Apply for x402 Foundation membership (Linux Foundation)	CEO + CTO	—	3.9
4.2	Build @gsdc/x402-facilitator (POST /verify · POST /settle · GET /supported)	Dev Lead	3–4 wk	3.9 · EIP-3009 live
4.3	Register GSDC as x402 payment asset (CAIP-2 chain ID)	Dev Lead + CTO	1 day	4.2
4.4	Build x402 Corridor Adapter module — Layer 5 plug-in	Dev Lead / Sandip	2 wk	4.2 v1.2
4.5	x402 session management (repeat caller identity)	Backend dev	1 wk	4.4
4.6	Security review: EIP-3009 front-running on PAYMENT-SIGNATURE header	Smart contract dev + auditor	1 wk	4.2
4.7	x402 micropayment daily volume cap in facilitator	Dev Lead	3 days	4.4
4.8	Extend Audit Layer with x402 event feed	Backend dev	1 wk	1.21 4.4
4.9	Testnet: AI agent makes x402 GSDC payment end-to-end	QA + Dev Lead	1 wk	4.2 4.4 4.5
4.10	Build @gsdc/mcp-server (4 tools wrapping Settlement Orchestrator)	Dev Lead	2 wk	4.2
4.11	Publish GSDC to x402 Bazaar discovery layer	Dev Lead	1 day	4.3 4.9
4.12	MCP server testnet: Claude agent calls get_quote and confirm_settlement	QA	1 wk	4.10
4.13	Mainnet go-live for x402 + MCP at constrained volume	CTO + ops	—	4.6 4.7 4.8 4.9 4.12

Phase 5 — GSDC Settlement Chain (Own appchain)

ID	Task	Owner	Effort	Dependency
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5.1	Gateway.fm RaaS contract signed	CTO + CEO	—	3.5
5.2	GSDC Chain local CDK testnet (Docker quickstart)	Sandip	1 day	5.1
5.3	GSDC Chain Sepolia testnet (Gateway.fm managed prover)	Gateway.fm + Sandip	1–2 wk	5.1
5.4	Deploy all settlement contracts to GSDC Chain testnet	Sandip	1 wk	5.3
5.5	Bridge contracts audit (CertiK — chain layer)	CTO + CertiK	2–3 wk	5.4
5.6	AggLayer registration + CAIP-2 chain ID registration	Dev Lead + Gateway.fm	1 wk	5.3
5.7	Block explorer (Blockscout self-hosted)	Sandip	1 day	5.3
5.8	Sequencer in-house migration (TGS runs own sequencer)	Sandip + ops	2 wk	volume justification
5.9	Prover in-house migration (TGS runs own prover on Hetzner EU)	Sandip + ops	2 wk	5.8
5.10	GSDC Chain mainnet go-live — migration from Ethereum mainnet	CTO + ops	—	5.5 5.8 5.9

5. Open Decisions — What Still Gates Build

All architecture and product decisions are now closed. The following items are non-engineering — legal, commercial, or board decisions. None block Phase 1 engineering. All should be resolved in parallel.

P	Item	Owner	Gates
P0	Pathway D OTC counterparty identification	CEO / COO	Required before pilot go-live. Identify domestic Brazilian-licensed VASP this month.
P0	Brazilian counsel IOF opinion	Legal / COO	Gates Pathway C engineering decision and off-ramp cost model viability.
P1	Board confirmation of 0.5/0.5/0.5 margin rates	COO / CEO	Required before any partner term sheet. Engineering uses configurable marginBps[3] — board decision changes config not code.
P1	LP-BR partner signed	CMO / CEO	Gates adapter config lock (2.7) and pilot go-live (3.1). Phase 1 builds without this.
P1	LP-HK partner signed	CMO / CEO	Gates adapter config lock (2.8) and pilot go-live (3.1). Phase 1 builds without this.
P2	Partner technical discovery (HashKey / FOMO Pay / StraitsX)	CMO / CEO	Informs post-pilot chain preference and corridor expansion. No Phase 1 impact.
P2	Polygon CDK lifecycle commitment from Polygon Labs	CTO	Written 5-year CDK enterprise roadmap commitment before signing with Gateway.fm.
P2	QuickPay codebase licensing (Uruguayan counsel)	Legal / CTO	Chain-of-title for audit. Does not block Solidity build.
P2	CertiKit quote on audit duration	CTO	Timing input for pilot go-live date. Request as soon as code freeze is visible.

6. Document Registry — Everything Produced

All documents produced across the full architecture session. Read in order for new team members.

Document	Version	Status	Purpose
CTO Architecture & Operations Guide	v4.0	done	This document. Master reference for architecture, components, roadmap, and todo list.
Master Roadmap & Todo List	v4.0	done	Phase-by-phase build plan with owners and dependencies. Embedded in this document as Section 4.
Settlement CTO Response to v2	v3.0	done	Closes all four open questions in the Settlement System v2 memo. Primary document for CEO/COO.
PartnerAdapter Interface Specification	v1.2 FINAL	done	SIGNED OFF. All 18 gaps resolved. Four cross-check fixes. Three-state compliance. complianceFields on settlement types.
Partner Platform PRD — Track A	v1.1 FINAL	done	SIGNED OFF. Multi-tenant. Corridor Profiles. BCB/HKMA compliance modules. 14 jurisdictional gaps resolved.
CTO Sign-Off Record	May 2026	done	Formal record of both sign-offs. Four fixes documented. Change control rules. What each person starts this week.
PartnerAdapter v1.2 Delta	v1.2 delta	done	Read alongside v1.1. Contains only changed types and new compliance decision tree section.
CTO Architecture Guide v3	v3.0	deferred	Previous version. Superseded by this document. Retain for audit history.

6.1 Reading order for new engineering team members

- 1. This document (CTO Architecture Guide v4) — start here for the full picture
- 2. Settlement CTO Response v3 — understand the six-step settlement flow and three-actor model
- 3. PartnerAdapter Interface Spec v1.1 — full interface spec with all types
- 4. PartnerAdapter v1.2 delta — the four changes applied at sign-off
- 5. Partner Platform PRD v1.1 — what is being built for the partner-facing layer
- 6. Sign-Off Record — change history and what everyone starts this week